



Digital Signal Intelligence

A New Information Substrate for Insurance

Abstract:

Digital Signal Intelligence (DSI) defines a new, externally observable information substrate for insurance, enabling deterministic, autonomous risk assessment, exposure estimation, loss propensity inference, and pricing at scale.

IP Statement:

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BEGIN



Digital Signal Intelligence

A New Information Substrate for Insurance



Learn about DSI

Explore topics such as:

- The Automation Myth
- The PageRank Precedent
- The Core Thesis



Interact with DSI

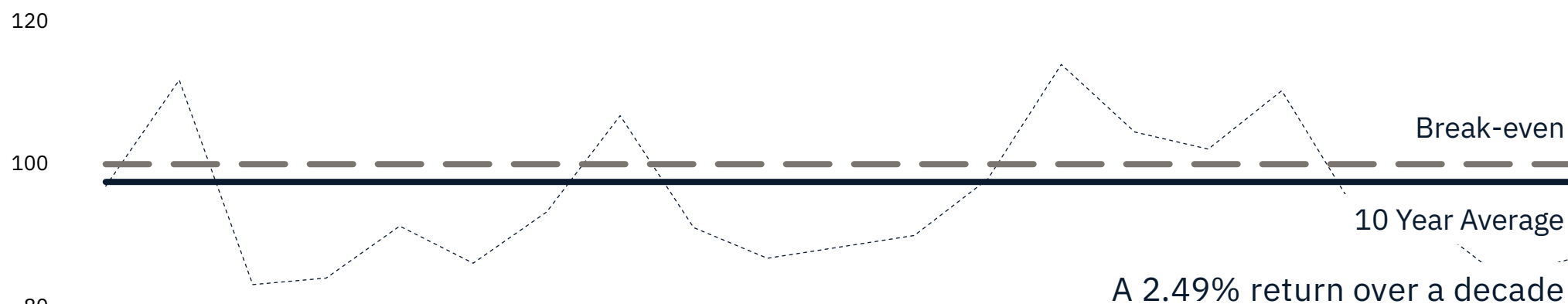
Experience the tool across sections such as:

- Human Oversight
- Portfolio Steering
- What Happens If

The Automation Myth

Decades of Poor Returns

Twenty years of Lloyd's combined-ratios show the same truth as every major market: insurance is flat-lining. Billions spent on process-modernisation and transformation have produced no structural improvement.



The core assumption is that traditional underwriting inputs can be automated.
This assumption is incorrect.
The inputs themselves are the constraint.

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The Information Substrate Problem

An Insurmountable Issue

Traditional Insurance inputs exist in infinite formats with no canonical schema nor deterministic structure.



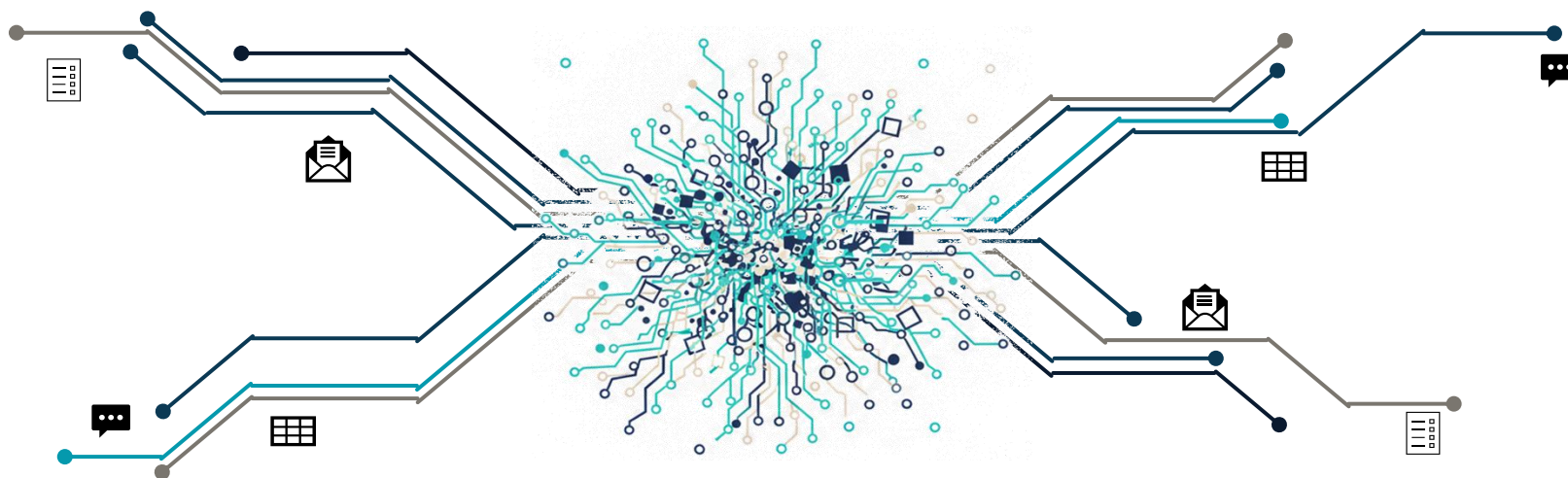
You'll recognise the maelstrom

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The Information Substrate Problem

An Insurmountable Issue



No extraction system, including modern or future AI models, can consistently transform these into the inputs required by zero-touch underwriting.

A system built on these inputs cannot achieve deterministic automation.

Autonomous underwriting requires the information substrate to change.

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Digital Signal Intelligence

The Core Thesis

Insurance underwriting requires understanding two properties of an organisation:

1. How well it is managed
2. How it behaves over time

Traditional underwriting attempts to infer these properties from self-reported documents, which are subjective, inconsistent, and non-verifiable.

DSI replaces them with observable digital signals.

All organisations operate within a digital ecosystem. Their participation in networks of clients, suppliers, partners, regulators, and certification bodies creates a citation network revealing trust and standing. Their digital trail of operational discipline provides a reliable proxy for maturity. Their hiring activity and investor communications reveal financial performance. Even the absence of expected signals is meaningful because well-run companies are predictable and uniform.

DSI uses these observable signals to create a deterministic, machine-readable foundation for underwriting without requiring self-reported input.

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Digital Signal Intelligence

Why Now?

DSI is not an experiment.

It is the latest evolution of a 30-year methodology that has revolutionised every major global industry by shifting from self-reported data to relational intelligence.

A REVOLUTIONARY INSIGHT

The Challenge	Search engines ranked pages by counting keywords. Result: Spam, page-gaming, and irrelevant results dominated search.
The Solution	Larry Page and Sergey Brin realised that a page's value isn't in its text, but in its citations, who links to it and the quality of those links and created PageRank.
The Result	Google captured 92% of the search market by prioritising the Digital Citation Network over content.

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GRAPH PROPAGATION

Between 1998 and 2020 digital citation algorithms have proven successful across every industry.

Consumer Finance (FICO)

Application: Credit scoring incorporates digital signals.

Result: Default prediction accuracy improved by 35-40%.

Recorded Future

Predict Cyber attacks via digital signals.

Threats are detected 6-8 weeks earlier than traditional methods.

Palantir and JP Morgan

Analyse digital signals to identify fraud.

Reduce false-positives by 50% and increase valid identification by 20%.

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The Birth of Modern AI

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THE BIRTH OF MODERN AI

Today, the principles established by PageRank have evolved into Modern AI.

The Evolution	Modern AI uses "Self-Attention." Just as PageRank assigns importance to web pages based on links, Transformers assign importance to words based on their relationship other in a sequence.
The Connection	A Transformer is essentially a dynamic, high-speed PageRank that happens in milliseconds.
The Link	Digital Signal Intelligence applies this "Attention" logic to insurance—treating 650B+ indexed signals as a neural network to identify risk.



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The Next Frontier

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THE NEXT FRONTIER

Beyond Transformers, the next frontier is World Models. While a Transformer reads context, a World Model simulates the environment's future.

DSI is the World Model for Insurance. It uses 650B+ signals to build a predictive simulation of an organisation's 'Hidden State,' allowing us to see a loss event in the digital substrate before it manifests in the physical world.

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The Bottom Line

Digital Signal Intelligence

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THE BOTTOM LINE

Not Experimental.

DSI's methodology was used by Amazon for recommendations (2002), LinkedIn for recruitment (2011), and modern AI applications from OpenAI, Google, and Anthropic being developed today. Insurance is the outlier.

Scale.

DSI taps into 1.1B websites and a thriving API economy used by the rest of the world already.

Reach.

Every entity uses the web. For example, 85% of mid-market companies have a meaningful web footprint.

Predictive Power.

From predicting cyber-attacks to predicting loan defaults, signal-based analysis is the global gold standard.

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Foundational Principles

Ensuring Consistent and Autonomous Operations

DSI operates consistently and autonomously. To ensure this, ten mandatory principles define the methodology:

1. **External Observability.**
Primary signals must be obtainable without cooperation.
2. **Machine Readability.**
All signals must be analysable by automated systems without human interpretation.
3. **Network Authority.**
Relationship citation patterns reveal quality
4. **Behavioural Inference.**
Observable behaviour trumps stated intentions.
5. **Absence as a Signal.**
Absence is a measurable indicator of risk.
6. **Structured Data Utilisation.**
Authoritative structured data is incorporated directly.
7. **Minimal Direct Inquiry.**
Direct questions are optional and constrained.
8. **Organisational-Level Assessment.**
Risk is assessed at the organisational level; parent behaviour is a proxy for underlying assets.
9. **Simplicity in Scoring.**
Decisions are visible and easily interrogated.
10. **Agentic Readiness.**
Models must be executable by autonomous systems.

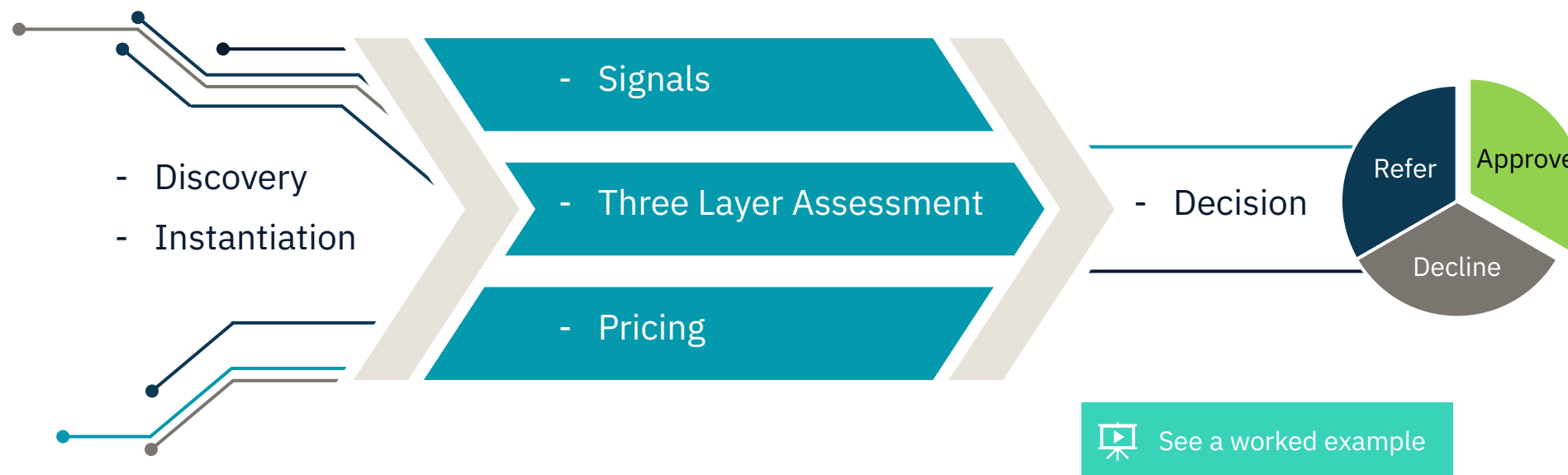
Ensuring consistency, objectivity, and agentic readiness in every assessment.

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End-to-End Model Architecture

Structured Workflow

Models convert raw signals into deterministic underwriting outputs through a structured workflow comprising six phases and fourteen discrete steps.



Critically, a post-bind signal loop will transform insurance from a static contract into a dynamic service.

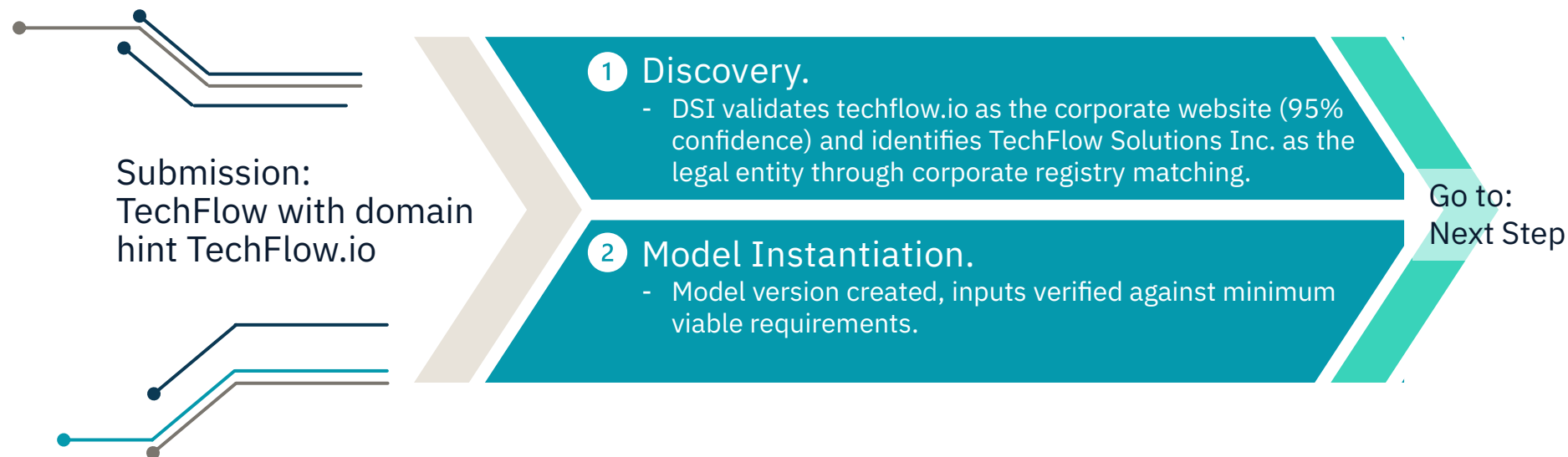


End-to-End Model Architecture

Structured Workflow

Worked Example

Exit example



End-to-End Model Architecture

Structured Workflow

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End-to-End Model Architecture

Structured Workflow

Worked Example

Exit example 

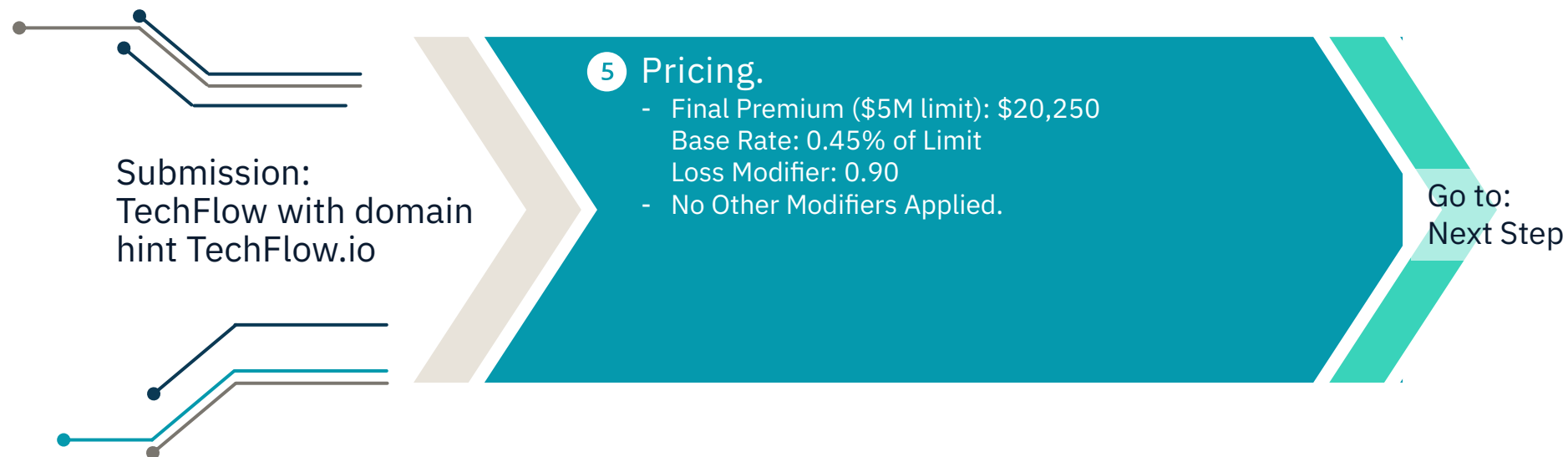


End-to-End Model Architecture

Structured Workflow

Worked Example

Exit example 

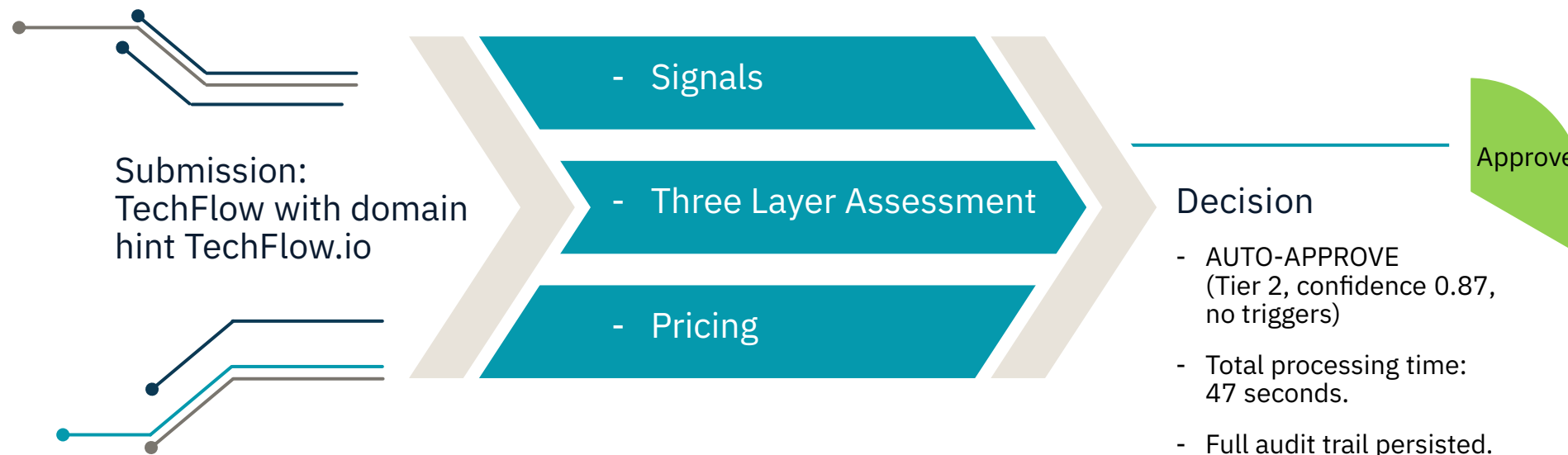


End-to-End Model Architecture

Structured Workflow

Worked Example

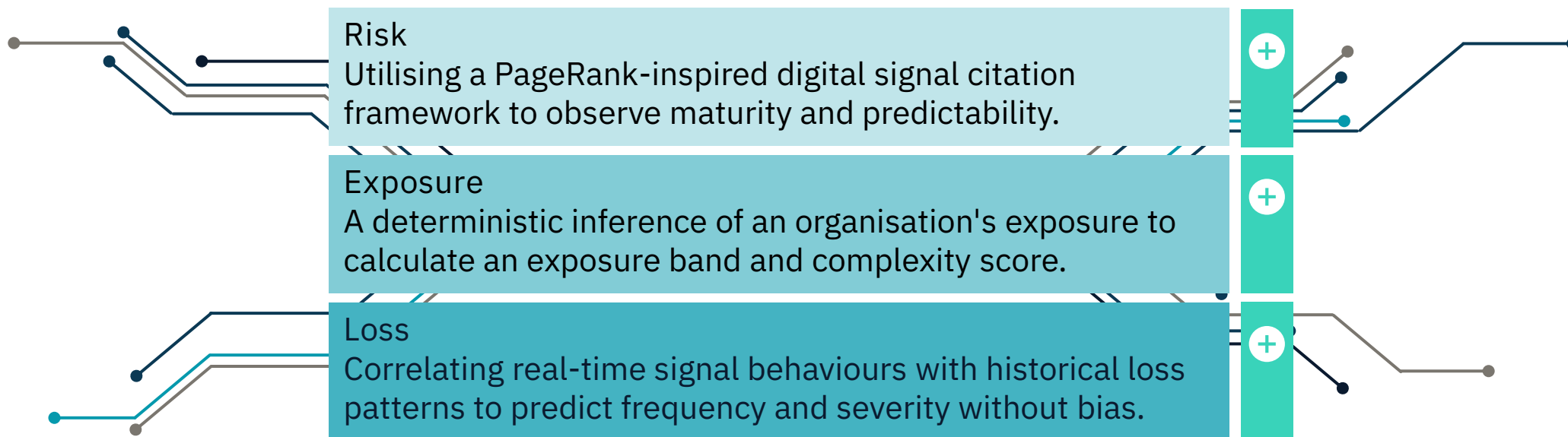
Exit example 



Three Layer Assessment

Decoupling Complexity for Absolute Clarity

Where traditional underwriting conflates three distinct analytical assessments into a single, subjective, aggregated assessment, DSI separates these into independent, measurable analytical dimensions.



Each dimension is assessed independently using the same underlying signal infrastructure, then combined into a unified pricing decision.

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Three Layer Assessment

Decoupling Complexity for Absolute Clarity

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Risk Analysis

The Risk Layer outputs six independent, risk focused dimensions:

1. **Composite Risk Score.**
Weighted, aggregated score between 0-1000.
2. **Risk Tier.**
Configurable categorisation, typically between 1-5.
3. **Confidence Score.**
Confidence score between 0-1 reflects data availability, source reliability, and signal freshness.
4. **Tier Overrides (if triggered).**
Deterministic rules that adjust the assigned tier when specific signal conditions are met.
5. **Referral Flags.**
Explicit indicators that require human review due to low confidence, conflicting signals, or triggered conditions.
6. **Deterministic Modifiers.**
Pre-defined, rule-based adjustments applied after tier assignment. No subjective judgement is permitted.

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Three Layer Assessment

Decoupling Complexity for Absolute Clarity

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Loss Correlation Layer

The Loss Correlation Layer outputs four independent, loss focused dimensions:

1. **Loss Propensity Score (0-100).**
Likelihood of loss frequency based on signal patterns.
2. **Severity Propensity Score (0-100).**
Likelihood of higher-severity outcomes.
3. **Loss Trend Band.**
Trend classification of propensity over time from very low to high.
4. **Cohort Assignment.**
Signal-derived peer groups that share loss characteristics.
Cohorts enable several capabilities: peer comparison for outlier detection, cohort-level loss expectancy for pricing, and migration tracking as entities move between cohorts over time.

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Three Layer Assessment

Decoupling Complexity for Absolute Clarity

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Exposure Shadow Layer

The Exposure Shadow Layer outputs two independent, exposure focused dimensions:

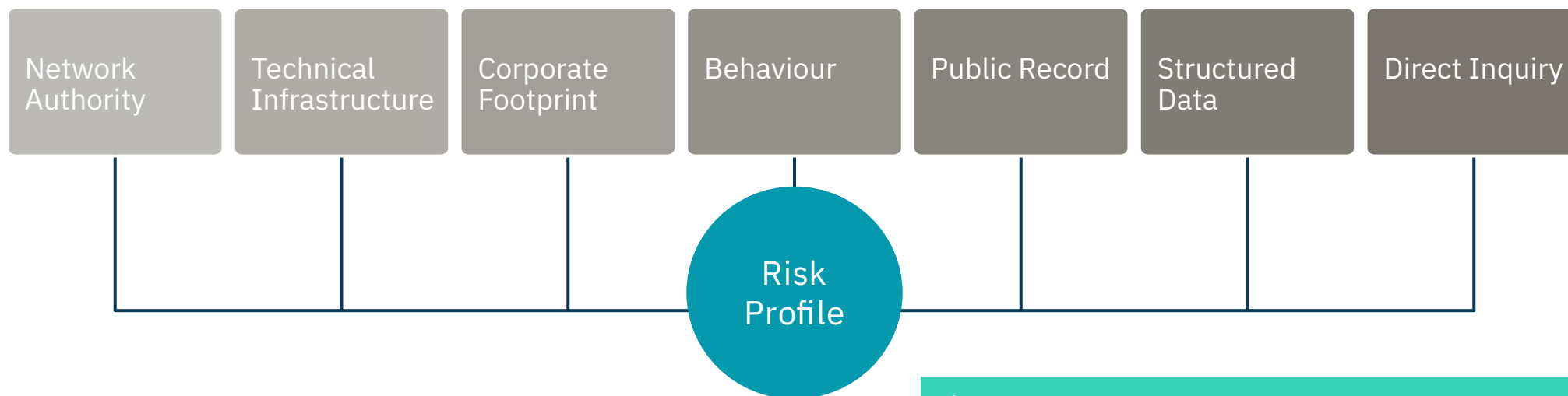
1. Exposure Band.
Classification of exposure magnitude.
2. Complexity Category.
Assessment of exposure structure complexity based on geographic dispersion, subsidiary count, technology heterogeneity, and regulatory footprint.

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Signal Architecture

Canonical Categorisation

DSI extracts signals from seven canonical categories that describe how an organisation behaves, operates, and participates in the digital ecosystem.



 See how signals would have caught historic losses

Signals are externally observable, machine-readable, and verifiable — enabling deterministic scoring without self-reported input.

Signal Architecture

Canonical Categorisation

Exit case-studies 

Retrospective Case-Studies

The Canonical categories are critical as they aggregate together multiple signals into logical cohorts. Each can be weighted individually and analysed to identify the features with a high-loss propensity.

Marine

Shadow Fleet (2022–2024)
Loss: 50+ incidents

DECLINE

Financial Institutions

Silicon Valley Bank (2023)
Loss: \$20B+ FDIC cost; \$200M+ D&O

DECLINE

Aerospace

Boeing 737 MAX (2018–2020)
Loss: \$18B+

REFER

Energy

Deepwater Horizon (2010)
Loss: \$3.4B insured; \$65B total

REFER + LOAD

Cyber

SolarWinds, December 2020
Loss: Global supply chain compromise

DECLINE

For each loss, the key signals we're prioritised for analysis and led to the submission being flagged for referral or declined.

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Competitive Differentiation

Why the Traditional Model Fails Where DSI Wins

DSI occupies a fundamentally different position in the underwriting landscape than existing signal-based tools.

Feature	Traditional	Digital Signal Intelligence
Input	Submissions (PDFs, Excel, Word)	Digital Ecosystem
Process	Manual, OCR, NLP	Deterministic Processing
Integrity	High "Manual Noise" ratio	Zero-Touch, Audit-Persistent
Adjustment	Subjective Overrides	Rule-Based Deterministic Modifiers
Scalability	Limited by manual "Stare & Compare"	Unlimited Agentic Readiness

This is a categorical distinction. DSI is not a better data feed—it is a complete underwriting substrate that happens to consume data feeds, including those from existing vendors, as inputs.

We don't make the old process faster; we make the old process obsolete.

While competitors fight for 10% efficiency gains in data entry, we are rebuilding the foundation.

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The Market

A \$1.8 Trillion Global Market — Ready for a New Substrate

DSI applies across the entire commercial insurance landscape because every meaningful organisation now leaves a measurable digital footprint.

This creates a universal, cross-coverage opportunity, with seven configured already.

- PI
- FI
- D&O
- Cyber
- Aerospace
- Marine
- Energy



With an AI Builder configured to help rapidly develop other coverages.

DSI is not a product for one line of business — it is a new underwriting substrate for the entire industry.

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Economic Impact

Improvements across the board

The following projections are based on model architecture and assume signal availability $\geq 80\%$ and straight-through processing rates $\geq 60\%$. Realisation timeline is estimated at 18–36 months from full deployment.

Expense Ratio  See the details

DSI targets a structural reduction in expense ratio of approximately 22 points.

Loss Ratio  See the details

DSI targets a structural improvement in loss ratios of 6-12 points.

Operational Metrics  See the details

DSI scales logarithmically as opposed to linearly by headcount.

On a \$500M premium book, these improvements represent \$130M–\$170M in improved profit.

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Economic Impact

Improvements across the board

The following projections are based on model architecture and assume signal availability $\geq 80\%$ and straight-through processing rates $\geq 60\%$. Realisation timeline is estimated at 18–36 months from full deployment.

Cohort	Metric	Traditional Model	DSI Target
Expense Ratio	Staff Costs	12%	4%
	Office and Infrastructure	8%	2%
	Administration	4%	1%
	Brokerage and Commissions	11%	6%
	Total	35%	13%
Loss Ratio	Loss Signal Correlation		4-8 points
	Continuous Calibration		2-4 points
	Total		6-12 points
Operational Metrics	Cost per submission	\$150–650	< \$10
	Straight-through rate	5–15%	60–80%
	Time to quote	3–10 days	< 60 seconds
	Scalability	Linear (headcount)	Logarithmic (compute)

Conclusion

The Future of Insurance is Agentic

Digital Signal Intelligence externally observable, machine-readable signals enable:

- Deterministic scoring without subjective interpretation
- Autonomous underwriting with deterministic guardrails at scale
- Exposure estimation without self-reported schedules
- Loss pattern inference from behavioural signals
- Continuous monitoring and early deterioration detection
- Global scalability without linear headcount growth

Insurers are publicly committing to reducing headcount; however, no traditional model can achieve this.

DSI is a structural replacement for the information substrate of underwriting, providing a viable path to radical operational efficiency and increased profitability.

As digital signals expand and calibration datasets grow, DSI becomes not only a new methodology for underwriting, but the foundation upon which the next generation of insurance carriers, built for Causal Simulation and real-time “World Model” risk prediction will be built.

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